

MAINTENANCE OPERATIONS CLIENT

KPI Catalogue

Maintenance Operations Analytics & Reporting Automation

Document Control	Detail
Engagement	Aug-Nov 2024
Role	Business Analyst + Data Analyst
Primary Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Purpose	Define and govern maintenance KPIs used across analysis, Power BI, DAX and testing
Status	Portfolio reconstruction grounded in confirmed engagement facts

KPI governance principle: Confirmed historical rules are identified explicitly. Where the exact historical calculation is not retained, the catalogue defines a portfolio-reconstructed specification rather than presenting an invented formula as historical fact.

1. Purpose

This KPI Catalogue is the measurement source of truth for the Maintenance Operations Client case study. It translates the business requirements and user stories into consistent analytical definitions that can be reused in the Power BI requirements, reconstructed DAX measures, operational analysis, dashboard validation and final report.

Critical distinction: The 92% overall completion rate is a finding surfaced by the analysis. It is not a performance improvement attributed to the project.

2. KPI Classification

Classification	Meaning
Confirmed historical rule	The calculation/business rule is retained from the engagement and may be stated directly.
Confirmed analytical area	The KPI/analysis was used, but its exact historical formula or mapping is not retained.
Portfolio-reconstructed specification	A defensible calculation is defined for rebuilding the case study and must be labeled as reconstructed.
Not retained	Exact historical values, thresholds, rankings or implementation details are unavailable and must not be invented.

3. KPI Summary

KPI / Analysis	Status	Business Purpose
Completion Rate	Confirmed rule	Measure the proportion of assigned maintenance tasks completed.
Completed Work Orders	Confirmed analytical area	Show completed maintenance workload.
Open Work Orders	Confirmed analytical area	Show work that remains open under the documented status mapping.
Pending Tasks	Confirmed analytical area	Show maintenance tasks categorized as pending.
Overdue Work Orders	Confirmed rule	Identify incomplete work past its due date.
Average Completion Time	Confirmed analytical area	Assess elapsed time for completed work.
Priority Breakdown	Confirmed analytical area	Understand work distribution by priority.
Team-wise Completion	Confirmed analytical area	Compare completion across teams.
Technician Performance	Confirmed analytical area	Analyze technician-level operational performance.
Backlog	Confirmed analytical area	Provide visibility into accumulated outstanding work.
High-Priority Pending Work	Confirmed analytical area	Surface critical work that remains pending.
Workload Imbalance	Confirmed finding/analysis area	Identify uneven workload distribution.
Recurring Categories	Confirmed finding/analysis area	Identify recurring maintenance categories.
Asset / Equipment Breakdown	Confirmed analytical area	Analyze maintenance activity by asset/equipment.

4. KPI-01 — Completion Rate

Evidence status: CONFIRMED HISTORICAL BUSINESS RULE

Attribute	Definition
Business question	What proportion of assigned maintenance tasks has been completed?
Formula	Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100
Numerator	Completed Maintenance Tasks
Denominator	Total Assigned Maintenance Tasks
Output	Percentage
Confirmed result	92% overall completion rate
Interpretation	92% of the eligible assigned maintenance tasks represented in the analyzed scope were classified as completed under the project logic.
Causal boundary	Do not state that the project increased completion to 92%.
Likely dimensions	Team, technician, priority, category, asset/equipment where supported by prepared data
BRD trace	BR-02, BR-03
User-story trace	US-01, US-02

5. KPI-02 — Completed Work Orders / Tasks

Evidence status: CONFIRMED ANALYTICAL AREA; EXACT HISTORICAL STATUS MAPPING NOT RETAINED

Attribute	Portfolio Specification
Business question	How much maintenance work is completed?
Reconstructed definition	Count of eligible maintenance records mapped to the standardized Completed status.
Required fields	Task / Work Order ID, Status
Control	Status values must be standardized before counting.
Data-quality rule	Duplicate eligible records must not inflate the count.
Historical-value boundary	No exact completed-work count is retained.
BRD trace	BR-03
User-story trace	US-02

6. KPI-03 — Open Work Orders

Evidence status: CONFIRMED ANALYTICAL AREA; RECONSTRUCTED CALCULATION SPECIFICATION

Attribute	Portfolio Specification
Business question	How much maintenance work remains open?
Reconstructed definition	Count of eligible records mapped to the documented Open status category.
Important boundary	Open and Pending must not be assumed to mean the same thing.
Required fields	Task / Work Order ID, Status
Control	Use an explicit status-mapping table in the portfolio rebuild.
Historical-value boundary	No exact historical open-work count is retained.
BRD trace	BR-04
User-story trace	US-02

7. KPI-04 — Pending Tasks

Evidence status: CONFIRMED ANALYTICAL AREA; RECONSTRUCTED CALCULATION SPECIFICATION

Attribute	Portfolio Specification
Business question	How much maintenance work is currently categorized as pending?
Reconstructed definition	Count of eligible records mapped to the documented Pending status category.
Required fields	Task / Work Order ID, Status
Control	Pending status mapping must be documented and kept distinct from Open unless source semantics support equivalence.
Historical-value boundary	No exact historical pending-task count is retained.
BRD trace	BR-05
User-story trace	US-02

8. KPI-05 — Overdue Work Orders

Evidence status: CONFIRMED HISTORICAL BUSINESS RULE

Attribute	Definition
Business question	Which maintenance items are past due and still incomplete?
Rule	Due Date < Current Date AND Status != Completed
Required fields	Task / Work Order ID, Due Date, Status
Date control	A valid Due Date is required for overdue classification.
Status control	Completed records are excluded under the confirmed rule.
Output	Count/list of overdue eligible maintenance records; rate may be reconstructed only if denominator is explicitly defined.
Historical-value boundary	No exact historical overdue count or percentage is retained.
BRD trace	BR-06
User-story trace	US-03

9. KPI-06 — Average Completion Time

Evidence status: CONFIRMED ANALYTICAL AREA; EXACT HISTORICAL FORMULA NOT RETAINED

Attribute	Portfolio Specification
Business question	How long does completed maintenance work take?
Reconstructed calculation	Average elapsed time between the documented start/reference date and Completion Date for eligible completed records.
Required fields	Created Date and/or documented start/reference date, Completion Date, Status
Critical boundary	The exact historical start-date convention is not retained; the rebuild must state which date is used.
Data-quality controls	Require valid dates; investigate negative or impossible intervals; do not impute unsupported dates.
Unit	Document explicitly in the rebuild (for example, days or hours).
Historical-value boundary	No exact historical average completion-time value is retained.
BRD trace	BR-07
User-story trace	US-04

10. KPI-07 — Priority Breakdown

Evidence status: CONFIRMED ANALYTICAL AREA

Attribute	Portfolio Specification
Business question	How is maintenance work distributed by priority?
Reconstructed calculation	Count and/or share of eligible maintenance records by standardized Priority.
Required fields	Task / Work Order ID, Priority
Control	Missing/unknown priorities remain identifiable; do not assign an unsupported priority.
Use	Overall workload analysis and high-priority exception visibility.
Historical-value boundary	Exact priority counts/shares are not retained.
BRD trace	BR-08
User-story trace	US-05

11. KPI-08 — Team-wise Completion

Evidence status: CONFIRMED ANALYTICAL AREA

Attribute	Portfolio Specification
Business question	How does maintenance completion differ across teams?
Reconstructed calculation	Apply the Completion Rate definition within each standardized Team filter context.
Required fields	Task / Work Order ID, Team, Status
Control	Team names must be standardized before comparison.
Interpretation boundary	Differences may be surfaced; causation must not be inferred without supporting evidence.
Historical-value boundary	Exact historical team percentages/rankings are not retained.
BRD trace	BR-09
User-story trace	US-06

12. KPI-09 — Technician Performance

Evidence status: CONFIRMED ANALYTICAL AREA; NO SINGLE HISTORICAL SCORE RETAINED

Attribute	Portfolio Specification
Business question	What operational patterns are visible at technician level?
Recommended reconstructed view	Use a set of measures such as assigned workload, completed work, completion rate, overdue work and completion time where supported.
Required fields	Technician plus the fields required by each underlying measure.
Control	Do not collapse technician performance into an invented composite score.
Missing attribution	Unknown/missing technician values must remain transparent.
Historical-value boundary	Exact historical technician rankings or scores are not retained.
BRD trace	BR-10
User-story trace	US-07

13. KPI-10 — Backlog

Evidence status: CONFIRMED ANALYTICAL AREA; EXACT HISTORICAL BACKLOG RULE NOT RETAINED

Attribute	Portfolio Specification
Business question	How much outstanding maintenance work has accumulated?
Reconstruction requirement	Define the eligible outstanding-work population explicitly before calculating backlog.
Potential inputs	Task / Work Order ID, Status, Due Date and other fields only where the documented rebuild rule requires them.
Critical boundary	Do not equate backlog automatically with overdue work.
Control	The portfolio definition must be documented in the KPI/DAX/testing artifacts before use.
Historical-value boundary	Exact historical backlog definition, threshold and count are not retained.
BRD trace	BR-11
User-story trace	US-08

14. KPI-11 — High-Priority Pending Work

Evidence status: CONFIRMED ANALYTICAL AREA

Attribute	Portfolio Specification
Business question	Which high-priority maintenance items remain pending?
Reconstructed calculation	Count/list of eligible records satisfying both the documented High Priority mapping and Pending status mapping.
Required fields	Task / Work Order ID, Priority, Status
Control	Both priority and status mappings must be explicit.
Historical-value boundary	No exact historical count or percentage is retained.
BRD trace	BR-13
User-story trace	US-05

15. KPI-12 — Workload Imbalance

Evidence status: CONFIRMED FINDING / ANALYTICAL AREA; EXACT HISTORICAL THRESHOLD NOT RETAINED

Attribute	Portfolio Specification
Business question	Is maintenance workload distributed unevenly across operational groups?
Reconstructed method	Compare documented workload measures across teams and/or technicians.
Possible workload measure	Assigned work volume; additional measures may be used only if defined in the rebuild.
Control	Do not invent a historical imbalance threshold.
Interpretation	Use as an analytical signal requiring operational context, not proof of underperformance.
Historical-value boundary	Exact imbalance percentages/thresholds are not retained.
BRD trace	BR-12
User-story trace	US-06, US-08

16. KPI-13 — Recurring Maintenance Categories

Evidence status: CONFIRMED FINDING / ANALYTICAL AREA

Attribute	Portfolio Specification
Business question	Which maintenance categories recur most often or show repeated activity?
Reconstructed method	Frequency and/or trend analysis by standardized Category.
Required fields	Task / Work Order ID, Category; dates where trend analysis is used.
Control	Category labels must be standardized before aggregation.
Interpretation boundary	Recurrence identifies patterns; it does not by itself establish root cause.
Historical-value boundary	Exact category frequencies/rankings are not retained.
BRD trace	BR-14
User-story trace	US-09

17. KPI-14 — Asset / Equipment Breakdown

Evidence status: CONFIRMED ANALYTICAL AREA

Attribute	Portfolio Specification
Business question	What maintenance activity and issue patterns are visible by asset/equipment?
Reconstructed method	Aggregate relevant maintenance measures by Asset / Equipment.
Required fields	Asset / Equipment plus fields required by the selected measures.
Possible views	Work volume, completed work, overdue work, categories and recurring activity where supported.
Interpretation boundary	Do not infer equipment failure causation from maintenance-record frequency alone.
Historical-value boundary	Exact historical asset rankings/counts are not retained.
BRD trace	BR-15
User-story trace	US-09

18. KPI Filter Dimensions

The following dimensions were present in the confirmed data landscape and can support KPI slicing where the prepared data is valid:

- Team
- Technician
- Asset / Equipment
- Status
- Priority
- Category
- Location
- Created Date
- Due Date
- Completion Date

Dimension control: Not every KPI must use every dimension. A filter should be enabled only where the relationship and business meaning are valid.

19. Data Quality Controls for KPIs

Data Issue	KPI Risk	Required Control
Duplicates	Inflated counts and distorted rates	Identify and control duplicates before aggregation.
Missing values	Incomplete attribution/classification	Handle explicitly; do not invent operational facts.
Inconsistent team names	Split team results	Standardize team representation.
Inconsistent date formats	Invalid overdue/time calculations	Parse and validate date fields.
Inconsistent status names	Incorrect completion/open/pending/overdue results	Use documented status mapping.
Blank records	Meaningless aggregation	Exclude/control non-actionable blank records.

20. KPI Validation Requirements

Validation ID	Validation Requirement
KVAL-01	Completion Rate must reconcile to Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100.
KVAL-02	The portfolio output may reproduce 92% only when the sanitized/reconstructed dataset is intentionally designed to support that confirmed finding or when the value is clearly presented as a retained engagement result.
KVAL-03	Overdue classification must satisfy Due Date < Current Date AND Status != Completed.
KVAL-04	Completed items must not appear as overdue under the confirmed rule.
KVAL-05	Open and Pending calculations must follow the documented rebuilt status mapping.
KVAL-06	Average Completion Time must exclude/control invalid date intervals according to the documented rebuild rule.
KVAL-07	Team-level completion must use the same completion definition as the overall KPI.
KVAL-08	High-Priority Pending Work must satisfy both documented priority and pending-status conditions.
KVAL-09	Backlog must use one explicit reconstructed definition consistently across DAX, visuals and tests.
KVAL-10	Equivalent measures must reconcile across dashboard pages under equivalent filters.
KVAL-11	No dashboard visual may present invented historical counts, rankings or percentages as client facts.
KVAL-12	Missing/unknown categories must remain visible or be handled according to documented data-quality rules.

21. KPI Traceability Matrix

KPI / Analysis	BRD	User Story	Downstream Artifact
Completion Rate	BR-02, BR-03	US-01, US-02	DAX, dashboard, testing
Completed/Open/Pending	BR-03 to BR-05	US-02	DAX, status analysis, testing
Overdue	BR-06	US-03	DAX, backlog/overdue analysis, testing
Average Completion Time	BR-07	US-04	DAX, work-order analysis, testing
Priority / High-Priority Pending	BR-08, BR-13	US-05	Priority analysis, dashboard
Team-wise Completion	BR-09	US-06	Performance analysis, dashboard
Technician Performance	BR-10	US-07	Performance analysis, dashboard
Backlog / Workload	BR-11, BR-12	US-08	Operational analysis, testing
Recurring Categories	BR-14	US-09	Operational analysis, dashboard
Asset / Equipment	BR-15	US-09	Operational analysis, dashboard

22. Portfolio DAX Boundary

The later Data Model & DAX Measures document may implement these KPI specifications as reproducible portfolio measures. Only the Completion Rate and Overdue definitions should be described as retained historical business rules. Other exact DAX expressions must be presented as reconstructed implementation logic.

Do not fabricate: Exact historical SQL, Python, Power Query or DAX code is not retained.

23. KPI Integrity Summary

Measurement chain: Prepared maintenance data -> controlled status/date/team standardization -> governed KPI definitions -> operational analysis -> Power BI measures -> validation -> management reporting.

This catalogue is intended to prevent metric drift across the portfolio. Later documents should reference these definitions rather than silently redefining completion, overdue, backlog, technician performance or other maintenance measures.